Name: Ganji Poornesh Naga Sai Hanuman

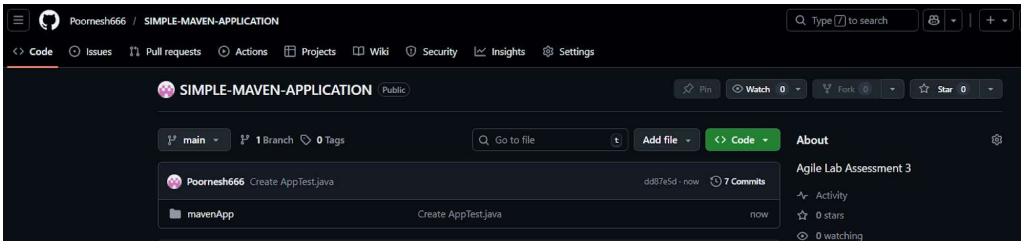
Regd. No: 23MIS0146

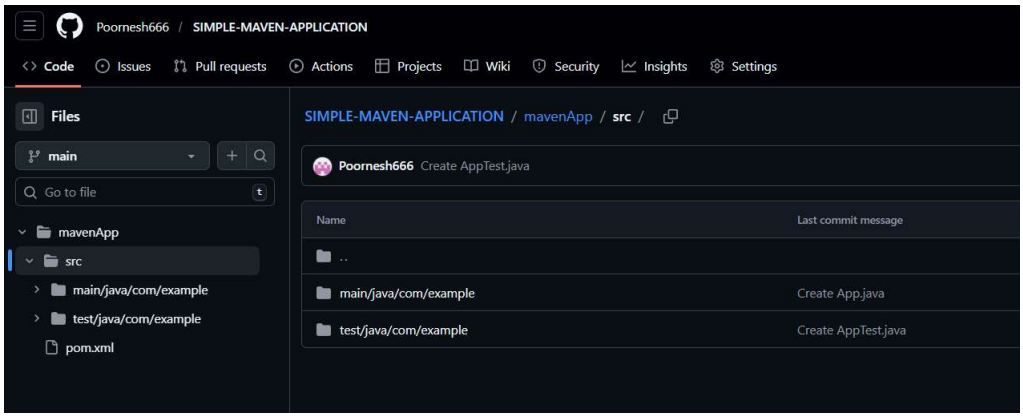
Agile & DevOps Lab Assignment 3 Document

Install Maven:

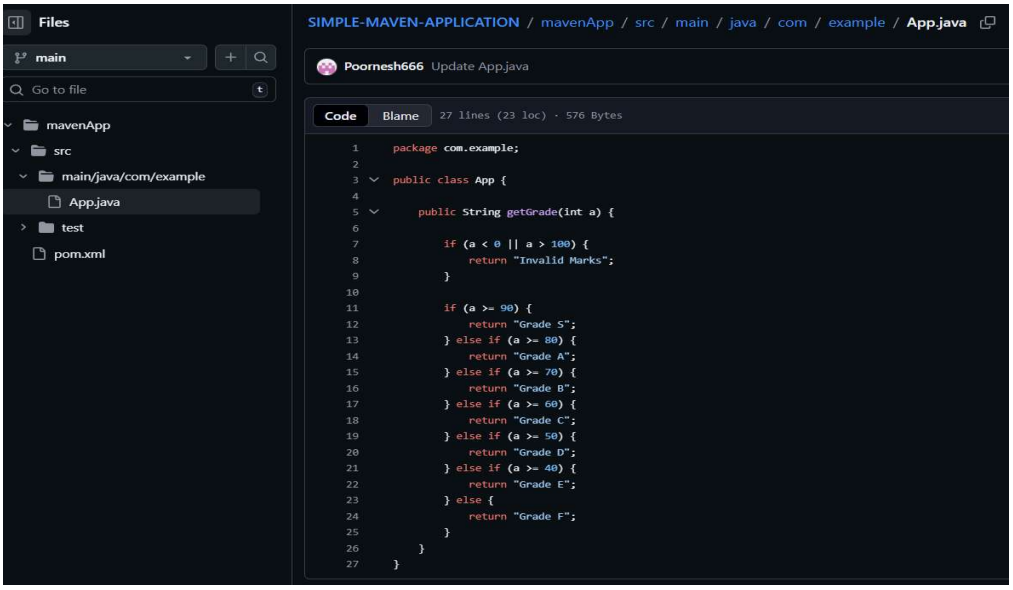
```
PS C:\Users\poorn> mvn --version
Apache Maven 3.9.11 (3e54c93a704957b63ee3494413a2b544fd3d825b)
Maven home: C:\ProgramData\chocolatey\lib\maven\apache-maven-3.9.11
Java version: 17.0.16, vendor: Amazon.com Inc., runtime: C:\Program Files\Amazon Corretto\jdk17.0.16_8
Default locale: en_IN, platform encoding: Cp1252
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"
PS C:\Users\poorn>
```

GITHUB CREATE A NEW REPOSITORY: SIMPLE-MAVEN-APPLICATION :





For Student Grade calculations: App.java:



AppTest.java:

Files

main

Go to file

- mavenApp
  - src
    - main
    - test/java/com/example
      - AppTest.java
  - pom.xml

SIMPLE-MAVEN-APPLICATION / mavenApp / src / test / java / com / example / AppTest.java

Code Blame 40 lines (32 loc) · 1.09 KB

```java
import org.junit.Test;
import static org.junit.Assert.*;

public class AppTest {

    @Test
    public void testGrades() {
        App app = new App();

        assertEquals("Grade S", app.getGrade(95));
        assertEquals("Grade A", app.getGrade(85));
        assertEquals("Grade B", app.getGrade(72));
        assertEquals("Grade C", app.getGrade(60));
        assertEquals("Grade D", app.getGrade(55));
        assertEquals("Grade E", app.getGrade(45));
        assertEquals("Grade F", app.getGrade(30));
    }

    @Test
    public void testEdgeCases() {
        App app = new App();

        assertEquals("Grade S", app.getGrade(100));
        assertEquals("Grade F", app.getGrade(0));
    }

    @Test
    public void testBoundaryValues() {
        App app = new App();

        assertEquals("Grade E", app.getGrade(40));
        assertEquals("Grade D", app.getGrade(50));
        assertEquals("Grade C", app.getGrade(60));
        assertEquals("Grade B", app.getGrade(70));
        assertEquals("Grade A", app.getGrade(80));
        assertEquals("Grade S", app.getGrade(90));
    }
```

Pom.xml:

Files

main

Go to file

- mavenApp
  - src
  - pom.xml

Poornesh666 Add Maven POM configuration for grading system

Code Blame 44 lines (33 loc) · 1.14 KB

```java
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>grading-system</artifactId>
    <version>1.0-SNAPSHOT</version>

    <name>Grading System</name>

    <properties>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
    </properties>

    <dependencies>

        <!-- JUnit 4 -->
        <dependency>
            <groupId>junit</groupId>
            <artifactId>junit</artifactId>
            <version>4.13.2</version>
            <scope>test</scope>
        </dependency>

    </dependencies>

    <build>
        <plugins>

            <!-- Surefire plugin to run tests -->
            <plugin>
```

Git clone in VS Code:

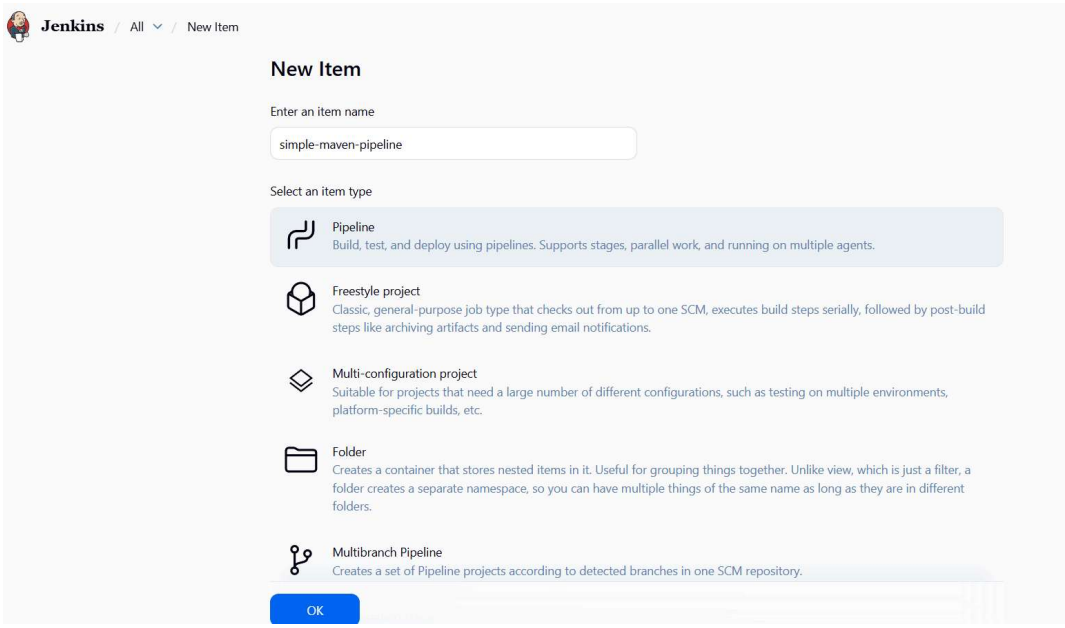
SIMPLE-MAVEN-APPLICATION

- mavenApp
  - src
    - main\java\com\example
      - App.java
    - test\java\com\example
      - AppTest.java
  - pom.xml

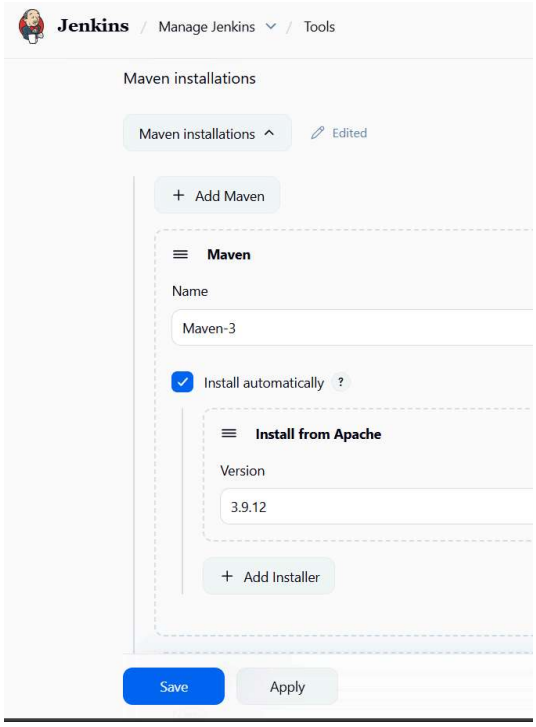
Output by mvn clean Test:

```
it4-3.2.5.jar (26 kB at 404 kB/s)
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.039 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  9.050 s
[INFO] Finished at: 2026-03-02T22:23:15+05:30
[INFO] ------------------------------------------------------------------------
```

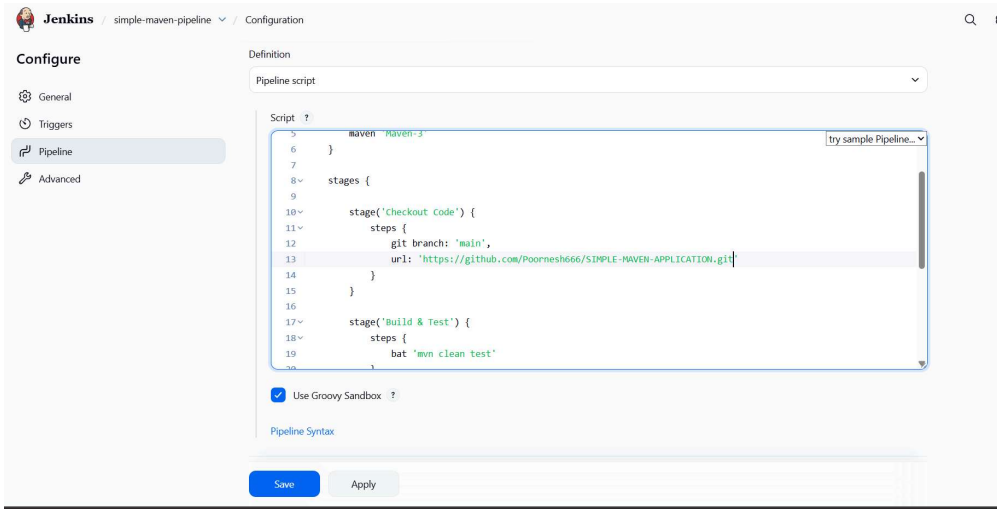
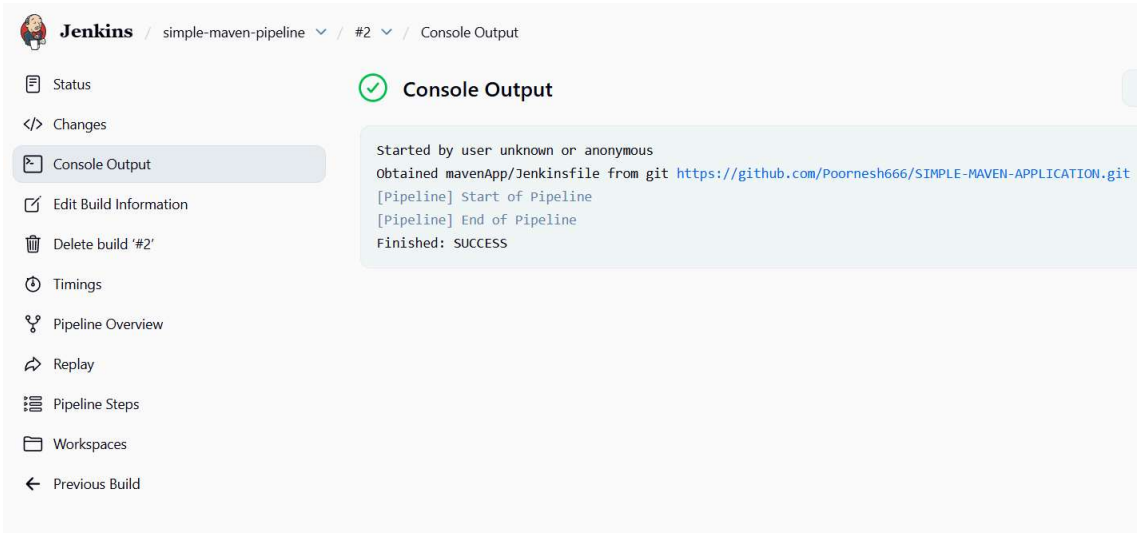
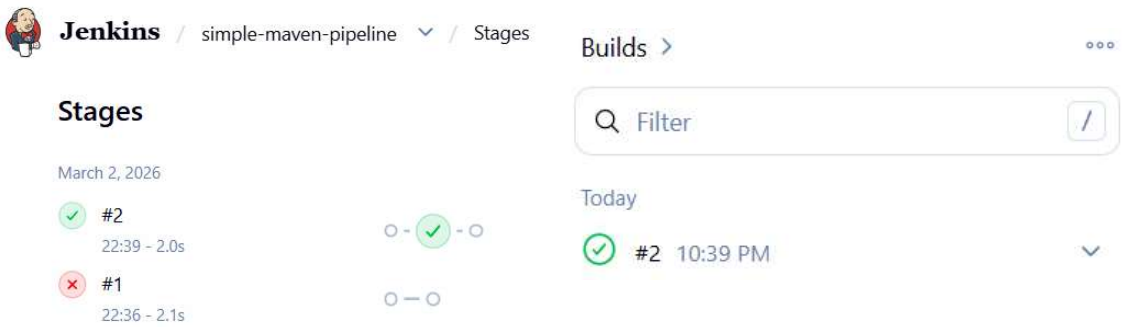
Move code in jenkins:



Maven Configuration in Jenkins:



Create Pipeline Script:



Successful Build Completion: